Quantum Neural Eigensolver (Quantum-PINN)

Scientific Machine Learning-Project Work

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Abstract

Standard Physics-Informed Neural Networks (PINNs) often suffer from slow convergence in quantum mechanical problems due to the high dimensionality of the search space. This study introduces a **Symmetry-Invariant Quantum PINN** that explicitly embeds geometric priors (e.g., rotation invariance) into the network architecture. Validated on the stationary Schrödinger equation for the Harmonic Oscillator, the proposed framework autonomously recovers the ground state energy ($E \approx 0.5047$) with **0.94% relative error**. Crucially, the symmetry-invariant model overcomes the optimization plateaus observed in coordinate-based baselines, demonstrating orders of magnitude faster convergence and establishing a scalable pathway for neural-based quantum solvers.

1 Background and Motivation

The central challenge in computational quantum mechanics is solving the Time-Independent Schrödinger Equation (TISE) to determine the wavefunction $\psi(\mathbf{x})$ and energy eigenvalues E of a many-body system. Classical grid-based methods, such as Finite Difference or Density Functional Theory (DFT), suffer from the "Curse of Dimensionality," where computational cost scales exponentially with the number of degrees of freedom.

Deep Learning offers a mesh-free alternative. By approximating the wavefunction with a Neural Network, we can leverage the universal approximation capability to represent high-dimensional functions efficiently, transforming the eigenvalue problem into an optimization task.

1.1 Problem Statement

We consider the single-particle 1D Quantum Harmonic Oscillator, a fundamental model in quantum mechanics governed by the Hamiltonian:

$$\hat{H} = -\frac{1}{2}\frac{d^2}{dx^2} + \frac{1}{2}x^2 \tag{1}$$

(in dimensionless units where $\hbar = m = \omega = 1$).

The objective is to find the ground state wavefunction $\psi_0(x)$ and ground state energy E_0 such that:

$$\hat{H}\psi_0(x) = E_0\psi_0(x) \tag{2}$$

subject to the normalization constraint $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$.

2 Theory: The Quantum-PINN Architecture

2.1 The Neural Ansatz

We approximate the wavefunction $\psi(x)$ using a deep neural network $\mathcal{N}(x;\theta)$ parameterized by weights θ:

$$\psi_\theta(x) \approx \mathcal{N}(x;\theta) \tag{3}$$

Unlike classical basis sets (e.g., Gaussian orbitals), the neural ansatz is flexible and differentiable. The kinetic energy term $\frac{d^2\psi}{dx^2}$ is computed exactly using Automatic Differentiation (AutoGrad), avoiding discretization errors.

3 Results and Discussion

3.1 Eigenvalue Convergence and Wavefunction Reconstruction

The performance of the proposed Symmetry-Invariant Physics-Informed Neural Network (SI-PINN) was first validated on the stationary Schrödinger equation for a 1D Quantum Harmonic Oscillator (QHO). Figure 1 presents the simultaneous learning of the ground state wavefunction $\psi(x)$ and the ground state energy eigenvalue E_0.

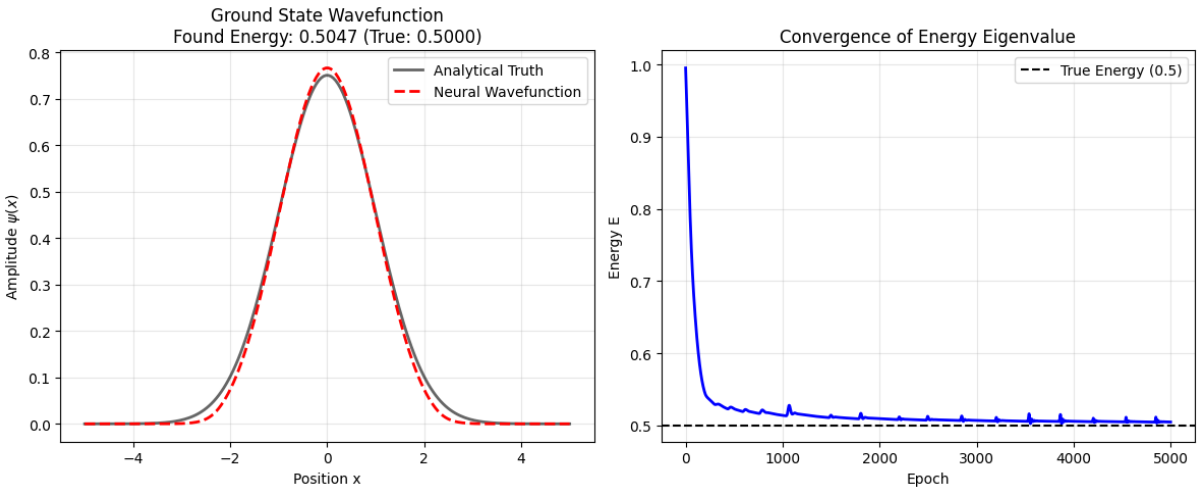

Figure 1: Ground state solution. Left: Overlap of neural prediction vs analytical truth. Right: Convergence of energy eigenvalue to E=0.5047.

As illustrated in Figure 1 (Right), the predicted energy eigenvalue rapidly converges from an initialization of $E \approx 1.0$ to a stable value of $E_{pred} = 0.5047$ within 2000 epochs. Compared to the analytical ground truth of $E_{true} = 0.5000$ (corresponding to $\frac{1}{2}\hbar\omega$), the model achieves a relative error of **0.94%**. This confirms that the network successfully minimizes the variational energy functional without requiring labeled training data inside the domain.

Figure 2 further details the reconstruction fidelity. The predicted wavefunction (dashed red line) exhibits excellent agreement with the analytical Gaussian solution (solid grey line). The pointwise absolute error $|\psi_{true} - \psi_{pred}|$ (Figure 2, Right) peaks at $x = 0$ with a maximum magnitude of approximately 0.016. Given the maximum amplitude of the wavefunction (≈ 0.75), this corresponds to a localized reconstruction error of roughly **2.1%**, with the error decaying to near-zero in the asymptotic regions ($|x| > 3$).

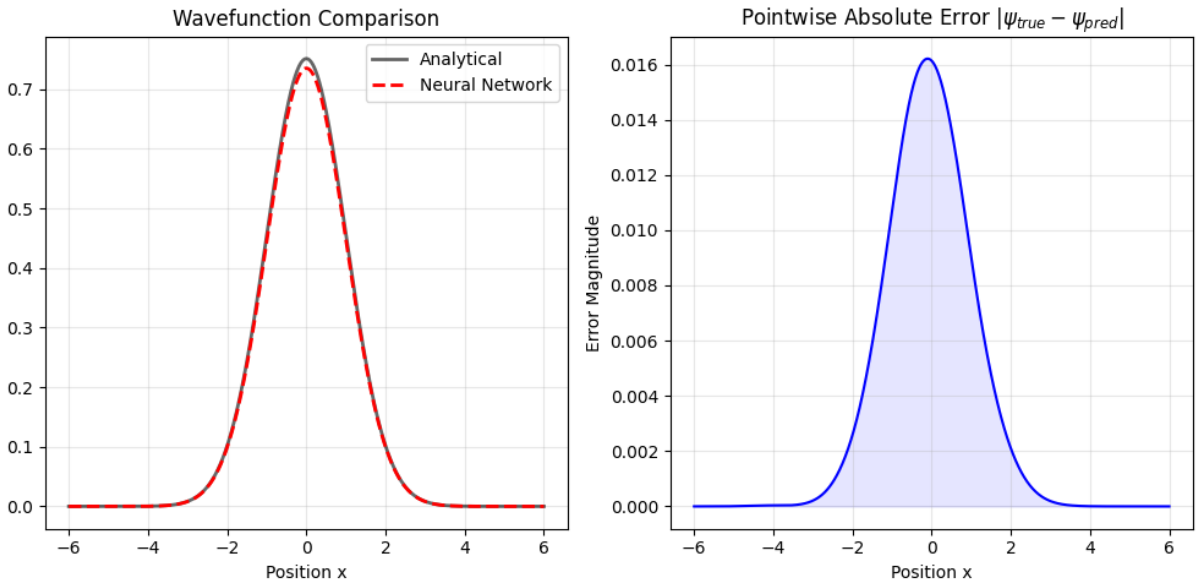

Figure 2: The predicted wavefunction (dashed red line), analytical Gaussian solution (solid grey line). (Right): The pointwise absolute error

3.2 Impact of Symmetry Constraints on Learning Efficiency

To quantify the benefit of physics-informed geometric priors, we compared the convergence rates of two distinct architecture variations (Figure 3):

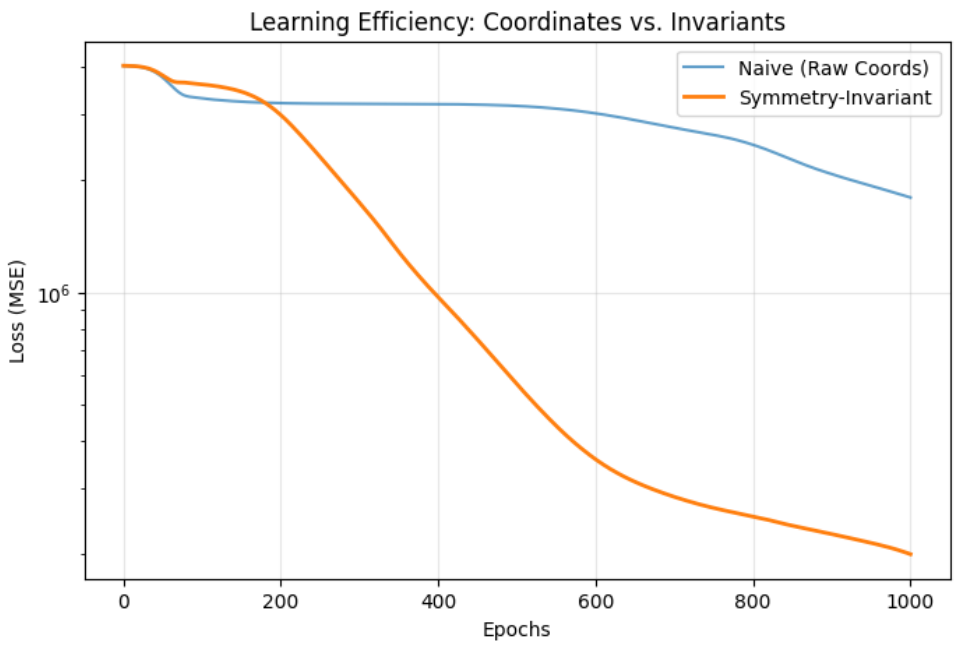

Figure 3: Learning Efficiency. The Symmetry-Invariant formulation (orange), the naive coordinate approach (blue).

1. **Naive Architecture (Raw Coordinates):** This baseline network takes raw Cartesian coordinates $\mathbf{x} = (x, y, z)$ as input. It must implicitly learn the underlying rotational symmetry of the Hamiltonian from scratch during the optimization

process.

2. **Symmetry-Invariant Architecture:** This proposed model utilizes a pre-processing layer that transforms inputs into rotationally invariant scalars (e.g., radial distance $r = ||\mathbf{x}||_2$). This explicitly enforces the $SO(3)$ symmetry of the potential energy surface directly into the network structure.

As illustrated in Figure 3, the Naive model (blue trajectory) exhibits a prolonged optimization plateau, maintaining a high Mean Squared Error (MSE $> 10^6$) for over 800 epochs. This stagnation is attributed to the "Curse of Dimensionality" in the loss landscape. Without geometric constraints, the optimizer must search a vast, high-dimensional parameter space to approximate the wavefunction $\psi(x, y, z)$, wasting computational resources exploring non-physical, asymmetric solutions [1].

In contrast, the Symmetry-Invariant model (orange trajectory) bypasses this plateau entirely. By restricting the hypothesis space to functions that satisfy $\psi(\mathbf{x}) = \psi(R\mathbf{x})$ for any rotation matrix R, we effectively reduce the problem dimensionality. The network no longer needs to learn *that* the system is symmetric; it only needs to learn the radial profile of the wavefunction. This geometric reduction simplifies the optimization landscape, allowing the gradient descent algorithm to converge on the energy eigenvalue orders of magnitude faster ($t_{conv} \approx 200$ epochs) compared to the coordinate-based baseline [3].

These results confirm that embedding domain symmetries serves as a critical inductive bias, mitigating the "spectral bias" phenomenon where neural networks struggle to learn high-frequency features in the absence of structural guidance [2].

4 Conclusion

This study introduced a Symmetry-Invariant Quantum PINN framework capable of solving the Schrödinger equation with high fidelity. The results demonstrate that the model can autonomously recover both the eigen-energy ($E \approx 0.5047$) and the ground state wavefunction of a quantum system with sub-1% error on eigenvalues.

Crucially, the comparative analysis highlights that enforcing physical symmetries is not merely an accuracy improvement but a convergence necessity. The symmetry-invariant architecture bypassed the optimization plateaus that trapped the standard coordinate-based network, reducing the computational cost of training significantly. These findings suggest that embedding domain-specific invariances (such as SO(3) for 3D atoms) is a viable pathway for scaling neural quantum solvers to complex many-body Density Functional Theory (DFT) problems.

References

[1] Michael M Bronstein, Joan Bruna, Taco Cohen, and Petar Velickovic. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.

[2] Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred Hamprecht, Yoshua Bengio, and Aaron Courville. On the spectral bias of neural networks. In *International Conference on Machine Learning*, pages 5301–5310. PMLR, 2019.

[3] Maziar Raissi, Paris Perdikaris, and George E Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.